Affine Schedule Response in a Synthetic Rotated-Surface-Code Fault Model

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Abstract

This note documents a prospective, fail-closed numerical mechanism experiment for an affine schedule response in a synthetic rotated-surface-code fault model. The experiment does not claim a threshold improvement, an asymptotic logical-error exponent change, a compiler-derived pulse model, a hardware advantage, or a reduction of real physical surface-code resources.

Claim Boundary

The ideal logical identity layer is inserted constructively by Stim. The schedule does not generate distinct physical unitary evolutions. Instead, it changes a frozen synthetic schedule-to-fault map: single-data and adjacent-pair fault probabilities are attached to the inserted ideal layer, and PyMatching is constructed from each arm detector error model.

Prospective Result

In the frozen v0.6.0 prospective cohort, specified and frozen before evaluation, constrained schedule optimization reduced the minimum tested one-sided Wilson-resolved distance from 11 to 9 at target logical failure probability pL = 0.025 on all three new prospective seeds. With syndrome rounds equal to distance, this corresponds to a 45.34% reduction in the active-coordinate qubit-round proxy, not a demonstrated reduction in calibrated hardware resources.

Reproducibility

The code repository for this artifact is https://github.com/papasop/quantum. The frozen protocol artifact is stored as results/v0.6.0/protocol.json and hashes to fec91e30001712f3d9ac84c0e45a6b70f2d5ae7189d3c9ac6d1096d47505cbf6 under the repository canonical JSON convention. The v0.6.0 reference report certificate hash is e4658480b6a4cb71caadfb6532453c4a31ba11f732dfaf36ffd66cc10921138c. Regenerate the full Monte Carlo report with the pinned requirements file.

Scientific Positioning

This is not an independent prediction validation: the optimization objective is built from the same synthetic schedule-to-fault probability model later injected into Stim. The Monte Carlo stage independently validates whether those probabilities, after syndrome extraction and decoding, cross the declared distance boundary.